

Exercise 1.1: Evaluating Model Performance

Name: Escalada, Lhei Matthew	Date: Jul 16, 2026
Course and Section: CPE 312-CPE31S3	Professor: Engr. Robin Valenzuela

1. Confusion Matrix

		Predicted	
		Spam	Non-spam
Actual	Spam	600 (True positive)	300 (False negative)
	Non-spam	100 (False positive)	9000 (True negative)



Compute the following:

A. Confusion Matrix

1. Accuracy = 96%

2. Misclassification Rate = 4%

3. True Positive Rate = 66.7%
4. False Positive Rate = 1.1%
5. True Negative Rate = 98.9%
6. Precision = 85.7%
7. Prevalence = 9%

B. Classification Report

1. Precision
2. Recall
3. F1 Score
4. Support

```

... Confusion Matrix:
TP=600, FP=100, FN=300, TN=9000
Accuracy: 0.96
Misclassification Rate: 0.040000000000000036
True Positive Rate (Recall): 0.6666666666666666
False Positive Rate: 0.01098901098901099
True Negative Rate (Specificity): 0.989010989010989
Precision: 0.8571428571428571
Prevalence: 0.09

Classification Report:

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	precision	recall	f1-score	support
Non-spam	0.97	0.99	0.98	9100
Spam	0.86	0.67	0.75	900
accuracy			0.96	10000
macro avg	0.91	0.83	0.86	10000
weighted avg	0.96	0.96	0.96	10000

Code:

Python

```
from sklearn.metrics import classification_report, accuracy_score

TP, FP, FN, TN = 600, 100, 300, 9000

y_true = ([1] * TP) + ([1] * FN) + ([0] * FP) + ([0] * TN)
y_pred = ([1] * TP) + ([0] * FN) + ([1] * FP) + ([0] * TN)

accuracy = (TP + TN) / (TP + TN + FP + FN)
misclassification_rate = 1 - accuracy
true_positive_rate = TP / (TP + FN)
false_positive_rate = FP / (FP + TN)
true_negative_rate = TN / (TN + FP)
precision = TP / (TP + FP)
prevalence = (TP + FN) / (TP + FN + FP + TN)

print("Confusion Matrix:")
print(f"TP={TP}, FP={FP}, FN={FN}, TN={TN}")
print("Accuracy:", accuracy)
print("Misclassification Rate:", misclassification_rate)
print("True Positive Rate (Recall):", true_positive_rate)
print("False Positive Rate:", false_positive_rate)
print("True Negative Rate (Specificity):", true_negative_rate)
print("Precision:", precision)
print("Prevalence:", prevalence)

report = classification_report(y_true, y_pred,
target_names=["Non-spam", "Spam"])
print("\nClassification Report:\n", report)
```